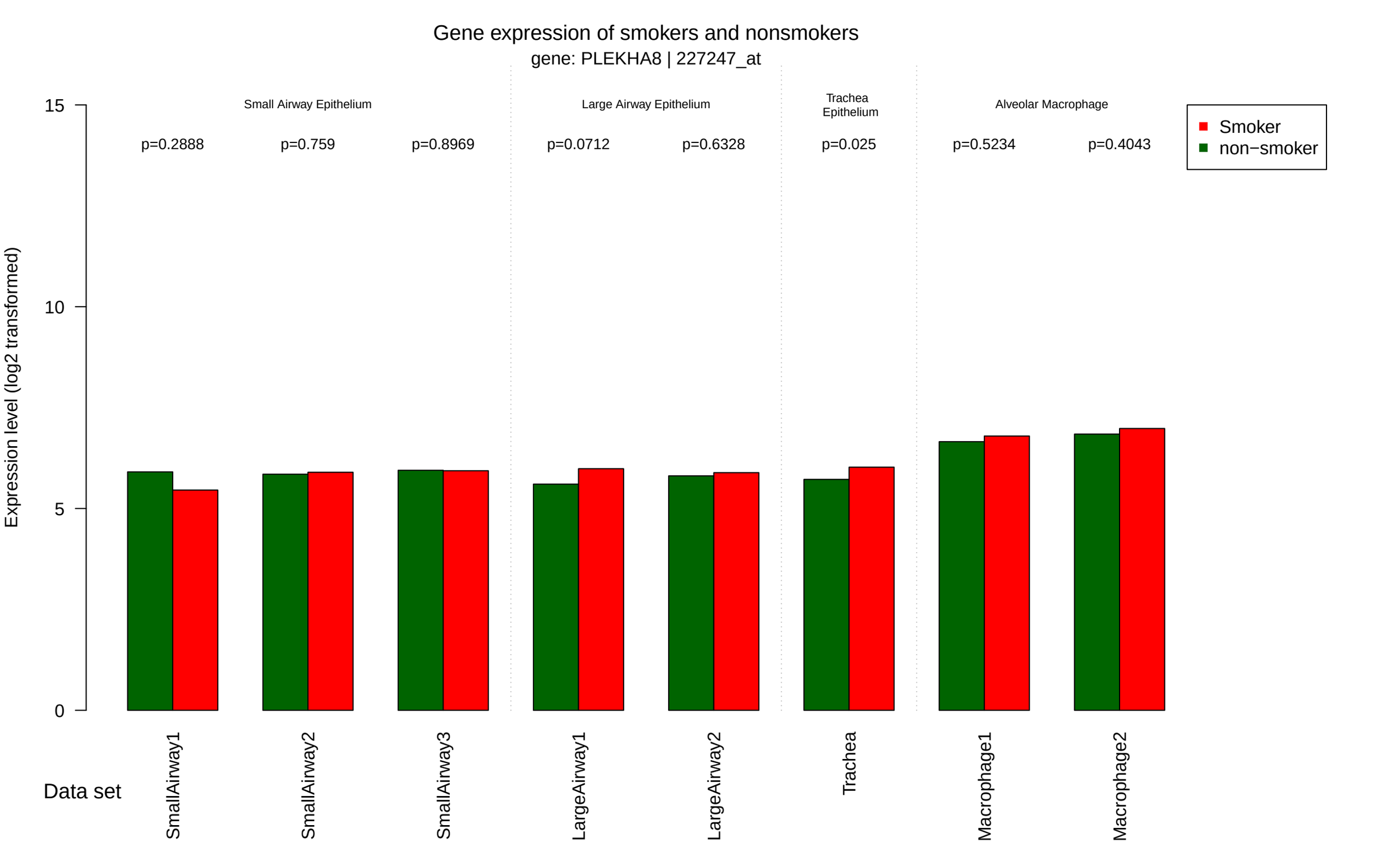
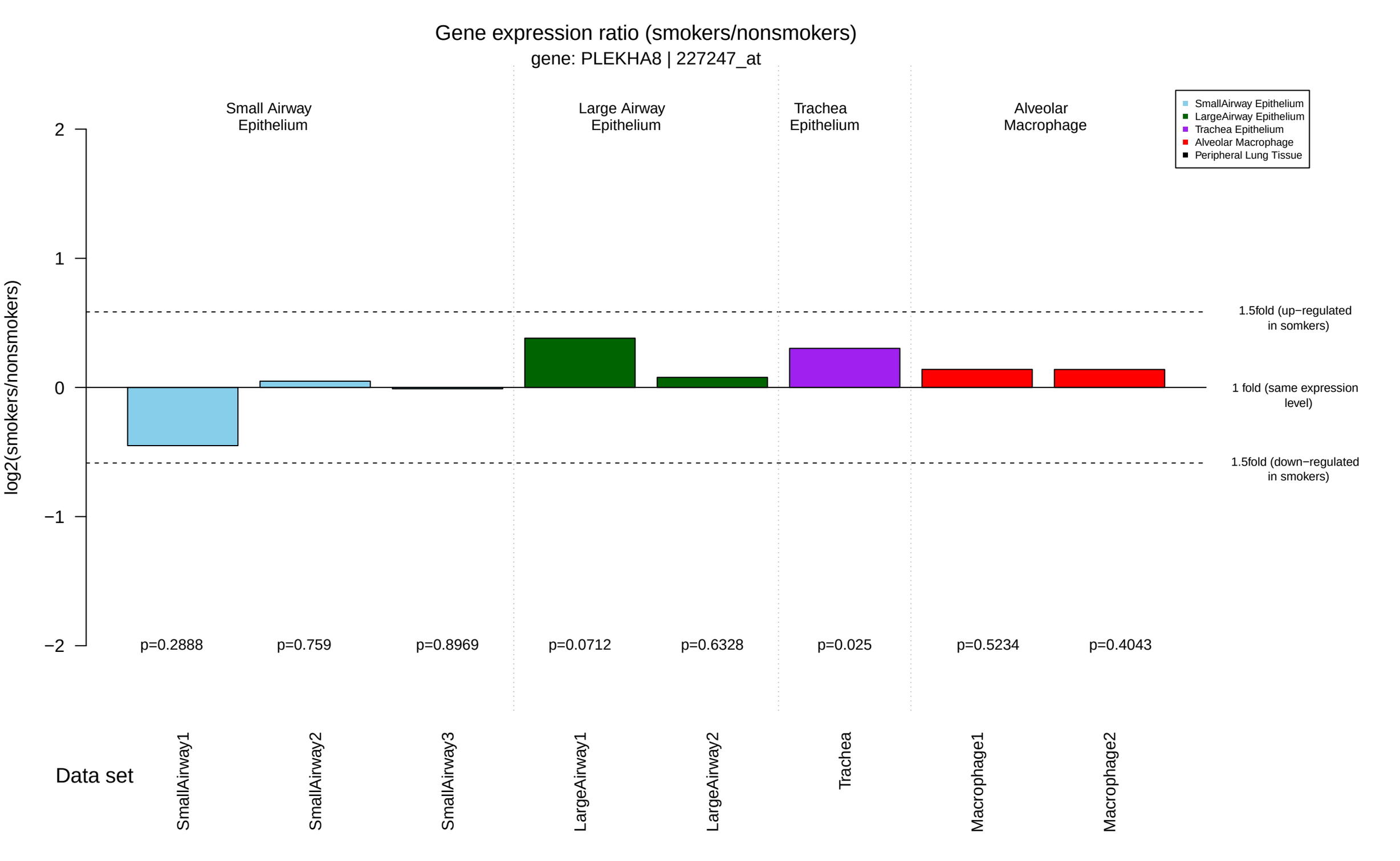
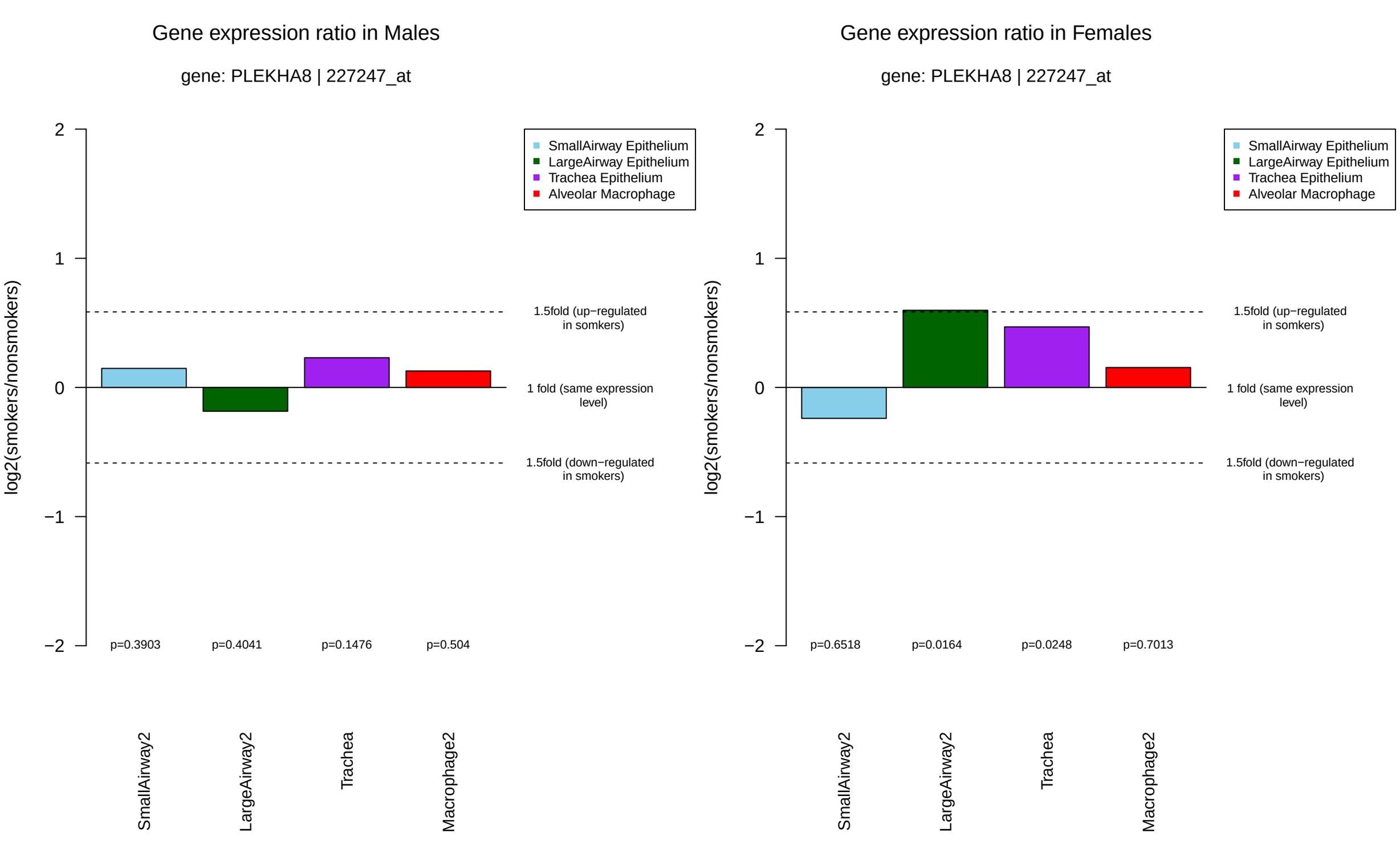
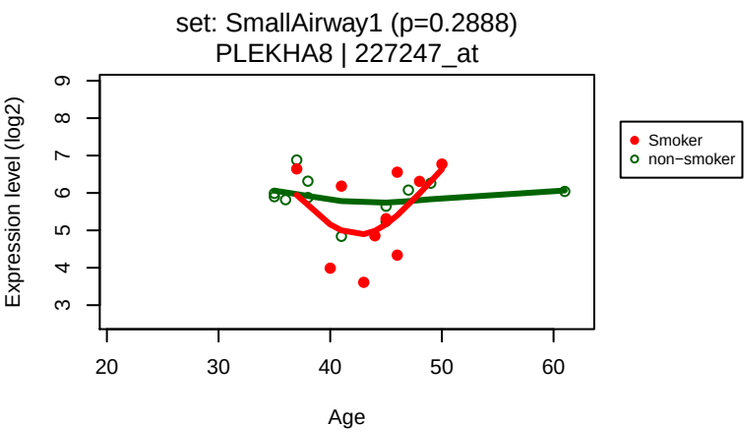
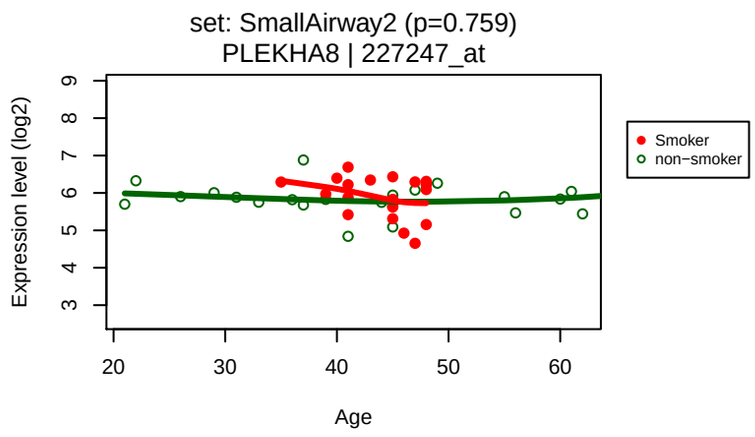
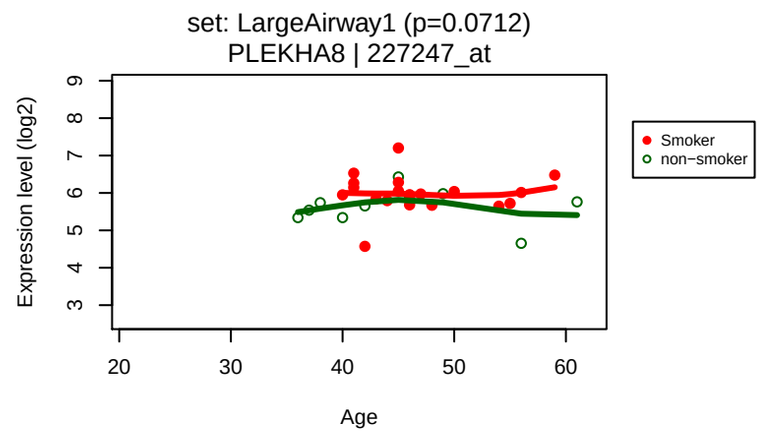

Gene expression of smokers and nonsmokers
gene: PLEKHA8 | 227247_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2888
p=0.759
p=0.8969
p=0.0712
p=0.6328
p=0.025
p=0.5234
p=0.4043
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

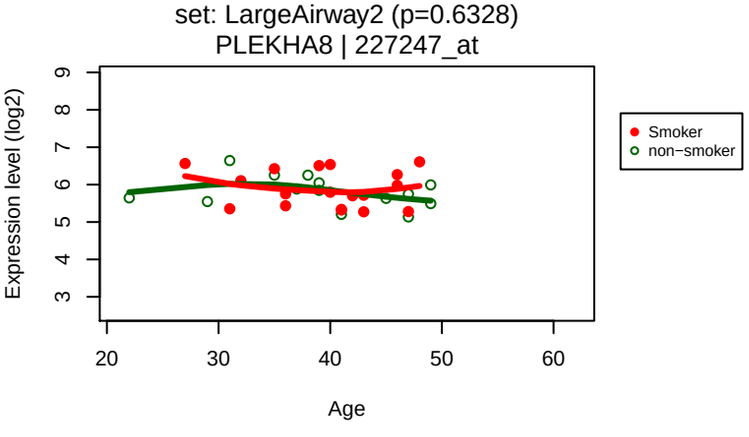

Gene expression ratio (smokers/nonsmokers)
gene: PLEKHA8 | 227247_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2888
p=0.759
p=0.8969
p=0.0712
p=0.6328
p=0.025
p=0.5234
p=0.4043
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

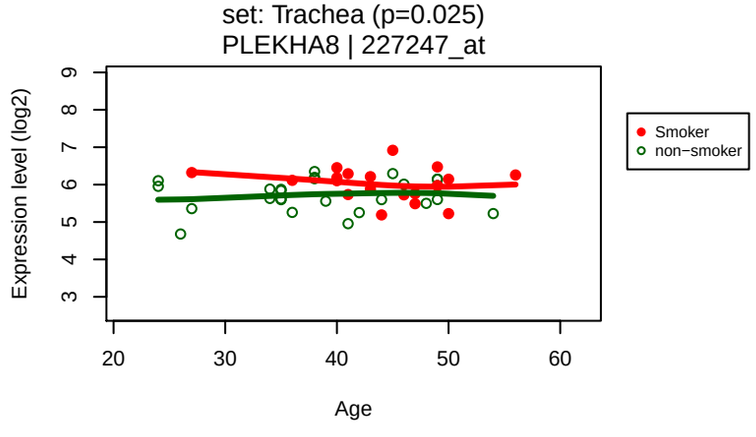

Gene expression ratio in Males
gene: PLEKHA8 | 227247_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3903
p=0.4041
p=0.1476
p=0.504
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PLEKHA8 | 227247_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6518
p=0.0164
p=0.0248
p=0.7013
SmallAirway2
LargeAirway2
Trachea
Macrophage2

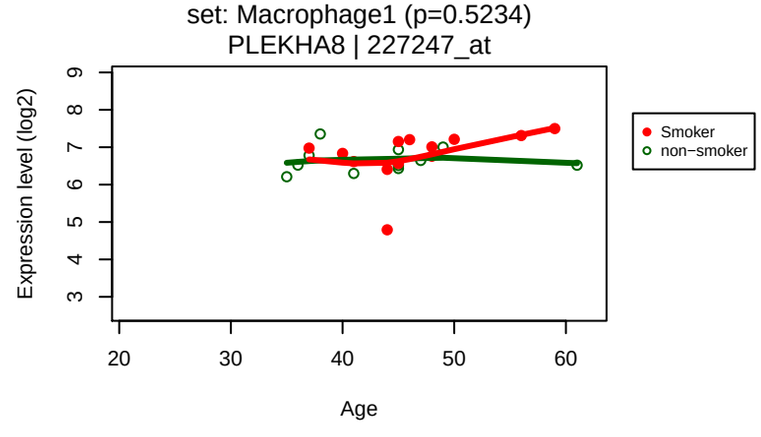

set: SmallAirway1 (p=0.2888)
PLEKHA8 | 227247_at
Expression level (log2)
Age
Smoker
non−smoker

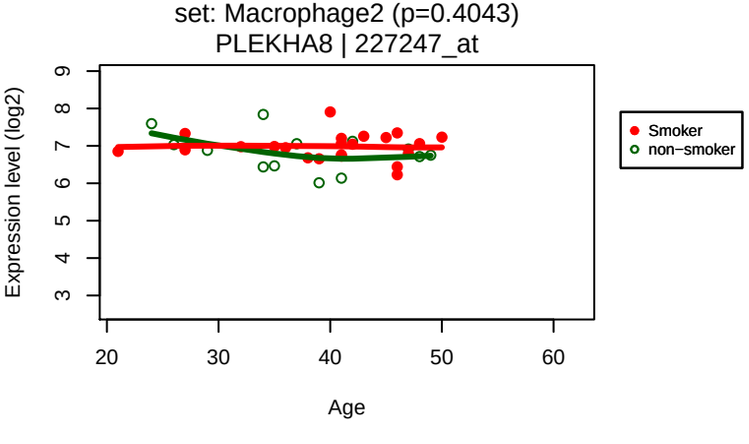

set: SmallAirway2 (p=0.759)
PLEKHA8 | 227247_at
Expression level (log2)
Age
Smoker
non−smoker

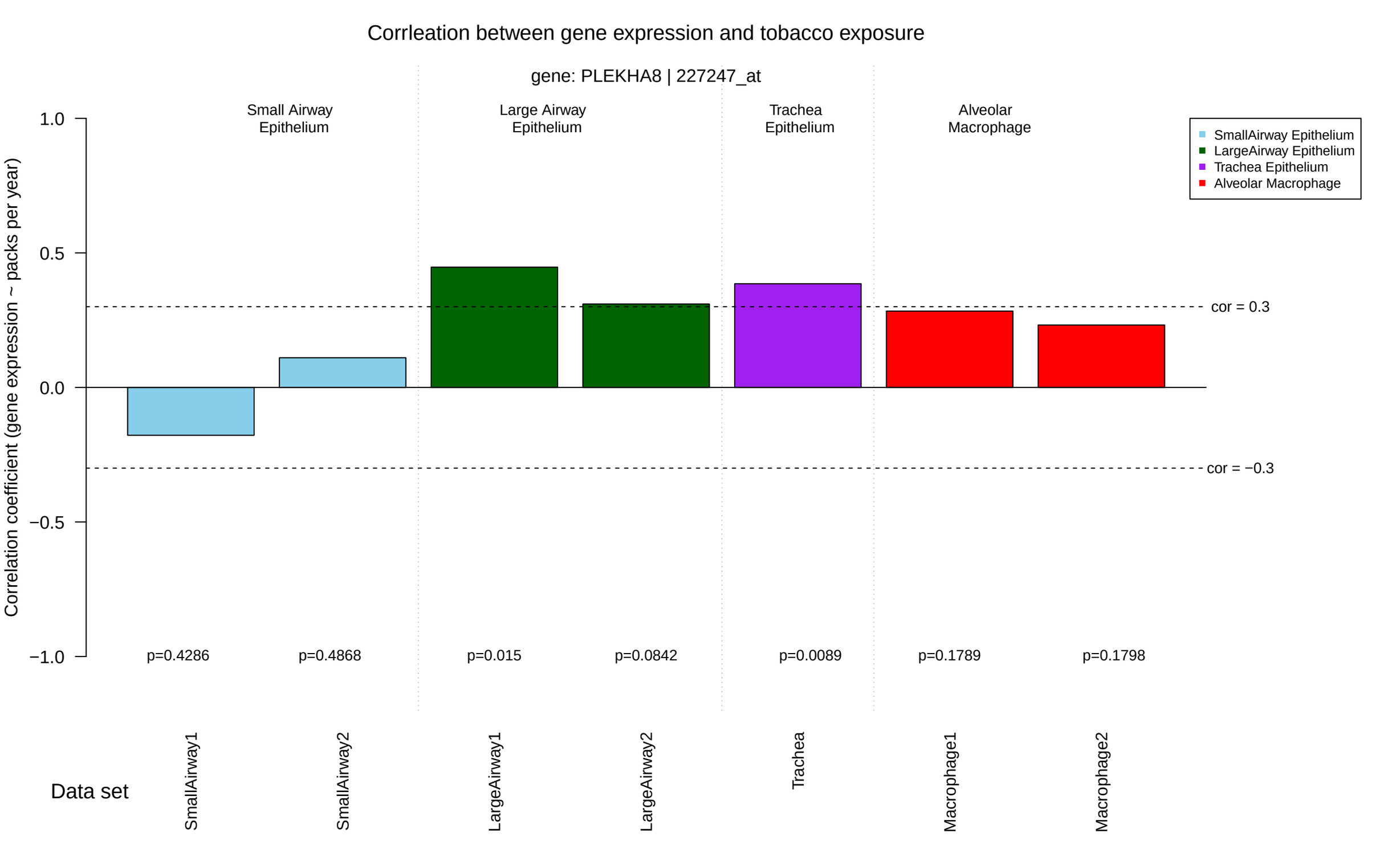

set: LargeAirway1 (p=0.0712)
PLEKHA8 | 227247_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6328)
PLEKHA8 | 227247_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.025)
PLEKHA8 | 227247_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5234)
PLEKHA8 | 227247_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4043)
PLEKHA8 | 227247_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PLEKHA8 | 227247_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4286
p=0.4868
p=0.015
p=0.0842
p=0.0089
p=0.1789
p=0.1798
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2